

GLOBAL HOUSING

MARKET ANALYSIS (2015–2024)

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OBJECTIVE

Analyze global housing market trends from 2015 to 2024, focusing on price indices, affordability, and their correlation with economic indicators like GDP growth and inflation.

DATASET

Global Housing Market Analysis (2015-2024) from Kaggle.

WHAT DO I DO?

I explore the dataset, clean it, perform exploratory analysis, visualize key trends, and derives insights into housing affordability and market drivers across selected countries.

```
#      Column      Non-Null Count  Dtype
---  -
0     Country      200 non-null    object
1     Year          200 non-null    int64
2     House Price Index  200 non-null    float64
3     Rent Index     200 non-null    float64
4     Affordability Ratio  200 non-null    float64
5     Mortgage Rate (%)  200 non-null    float64
6     Inflation Rate (%)  200 non-null    float64
7     GDP Growth (%)    200 non-null    float64
8     Population Growth (%) 200 non-null    float64
9     Urbanization Rate (%) 200 non-null    float64
10    Construction Index  200 non-null    float64
dtypes: float64(9), int64(1), object(1)
memory usage: 17.3+ KB
```

```
--- Missing Values Check ---
Country      0
Year         0
House Price Index  0
Rent Index   0
Affordability Ratio  0
Mortgage Rate (%)  0
Inflation Rate (%)  0
GDP Growth (%)  0
Population Growth (%) 0
Urbanization Rate (%) 0
Construction Index  0
dtype: int64
```

For example: Urbanization Rate (%) to Urbanization_Rate_Perc (to reduce chances of writing the wrong name during analysis at any point)

```
--- Columns Renamed ---  
Index(['Country', 'Year', 'House Price Index', 'Rent Index',  
      'Affordability Ratio', 'Mortgage_Rate_Perc', 'Inflation_Rate_Perc',  
      'GDP_Growth_Perc', 'Population_Growth_Perc', 'Urbanization_Rate_Perc',  
      'Construction Index'],  
      dtype='object')
```

Post Cleaning Analysis:

Minimum values for percentage columns appear reasonable (no negative mortgage rates observed).

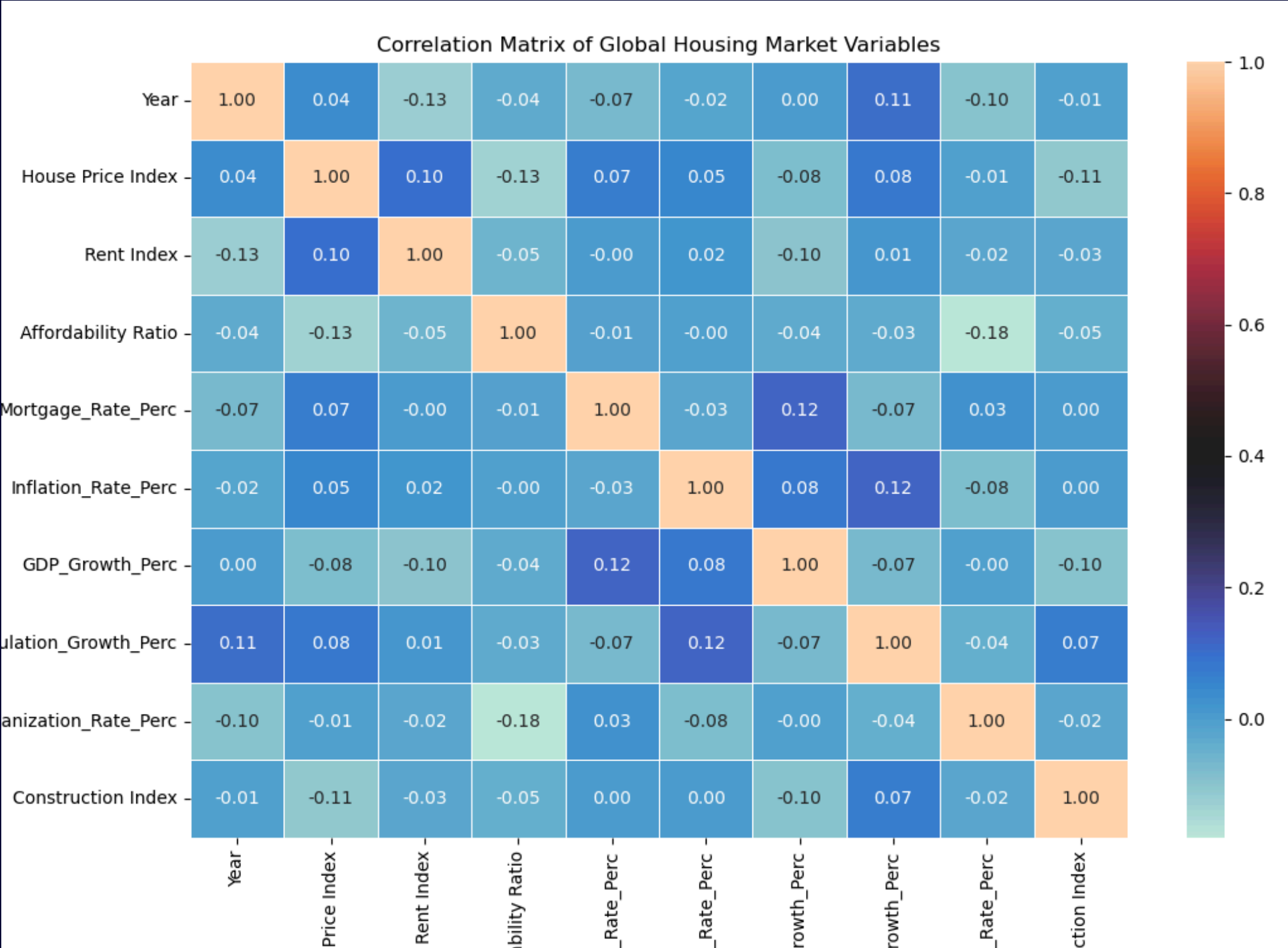
--- Summary Statistics (Post-Cleaning) ---				
	Year	House Price Index	Rent Index	Affordability Ratio \
count	200.000000	200.000000	200.000000	200.000000
mean	2019.500000	130.381022	83.048370	7.237768
std	2.879489	28.752229	21.439858	2.576085
min	2015.000000	80.552212	50.354311	3.041688
25%	2017.000000	104.142562	60.466671	5.034207
50%	2019.500000	129.193653	83.721711	7.375697
75%	2022.000000	157.127098	100.604665	9.276196
max	2024.000000	179.971767	119.855388	11.879671
	Mortgage_Rate_Perc	Inflation_Rate_Perc	GDP_Growth_Perc \	
count	200.000000	200.000000	200.000000	
mean	4.150621	3.649756	2.133722	
std	1.380222	1.881938	2.413270	
min	1.537814	0.532110	-1.921833	
25%	3.045278	1.939199	-0.095628	
50%	4.329643	3.664625	2.307548	
75%	5.217706	5.261720	4.272776	
max	6.485623	6.912349	5.958931	
	Population_Growth_Perc	Urbanization_Rate_Perc	Construction Index	
count	200.000000	200.000000	200.000000	
mean	0.722806	74.769891	111.204257	
std	1.035691	8.734246	24.309473	
min	-0.961391	60.172760	70.968678	
25%	-0.183320	66.923842	90.182198	
50%	0.722372	75.100060	110.593807	
75%	1.621317	82.677430	133.780445	
max	2.497948	89.788944	149.735748	

Wrangling+Cleaning: (Summary)

- **Missing Values:** No missing values were present.
 - **Column Renaming:** Columns with '%' were renamed to use '_Perc' for easier coding. This was done for standardization.
 - **Duplicates:** No duplicate rows were found.
 - **Type Conversion:** 'Year' kept as integer type. No other type conversions deemed necessary.
 - **Validity:** Basic statistics confirm reasonable ranges for numeric data.
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Correlation Heatmap

No strong correlation was found.



- 5.1.2 House Price Index Trends (Faceted Line Plot) – .html file**
- 5.2 Average House Price Index by Country (Bar Plot) – .html file**
- 5.3 Mortgage Rate Distribution (Histogram) – .html file**
- 5.4 House Price Index vs. Rent Index (Scatter Plot) – .html file**
- 5.5 Affordability Ratio Over Time (Line Plot) – .html file**
- 5.6 Average House Price Index by Country (Geo Map) – .html file**
- 6.1 Change in House Price Index (2015 vs 2024) – Bar Chart – .html file**
- 6.2 House Price Index vs. Affordability Ratio (2024) – Scatter Plot – .html file**
- 6.3 Average Affordability Ratio by Country (Bar Plot) – .html file**
- 6.4 Affordability Change vs. Average GDP Growth (Scatter Plot) – .html file**
- 8. Optional Research Question Analysis – .html file**

5.1.1 to 6.4 Findings

Affordability Changes: Housing affordability trends varied significantly across countries between 2015 and 2024. While the global average showed some fluctuation, individual countries experienced substantial improvements (e.g., Germany, Netherlands) or worsening (e.g., Canada, Russia, South Korea, UK). The average affordability ratio across all countries remained challenging, generally above 6.

Critical Affordability Issues: Based on the *average* ratio over the period, Russia, South Africa, Sweden, and the UK appear to face the most critical affordability issues among the countries studied. In the final year (2024), South Korea and South Africa stand out with poor affordability despite not having the highest HPI.

Economic Factors:

GDP growth did not show a consistent correlation with improving affordability across all countries. Several nations (Italy, Russia, South Korea) experienced above-median growth alongside worsening affordability, indicating other factors are at play.

- **Dataset Limitations:** The analysis is based on the 20 countries included in the dataset; findings may not represent all global regions. The indices used (HPI, Rent Index) are relative and don't represent absolute prices. The affordability ratio calculation method isn't specified in the source data description and likely doesn't account for regional income variations within countries. Data covers only up to 2024.
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Optimal Question Findings

- **Data Filtered:** The analysis focuses on the average trends across all included countries for the years 2019 through 2022, encompassing the core COVID-19 pandemic period.
 - **Visualization:** The line plots above show the trajectory of the global average for the House Price Index (HPI), Affordability Ratio, and Mortgage Rate during these specific years based on the dataset.
 - **Observations:**
 - **House Price Index:** The average global HPI increased steadily from 2019 (avg. 123.1) through 2020 (avg. 132.3) and peaked in 2021 (avg. 142.1). However, there was a significant drop in the average HPI in 2022 (avg. 120.3), bringing it below the 2019 level.
 - **Affordability Ratio:** Interestingly, despite the HPI rise initially, the average global Affordability Ratio showed an improvement (decreased) consistently from 2019 (avg. 7.98) to 2021 (avg. 7.44) and remained relatively stable at this improved level in 2022 (avg. 7.43). This suggests factors like income changes or falling mortgage rates might have offset the rising house prices on average during 2019–2021.
 - **Mortgage Rate:** The average Mortgage Rate decreased from 2019 (avg. 4.20%) to its lowest point in 2021 (avg. 3.49%). This period of lower rates coincided with the HPI increase. In 2022, the average mortgage rate saw a sharp increase back up to 4.14%, coinciding with the drop observed in the average HPI for that year.
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Thank You
